

### *Asynchronous perception thread (15 fps)*

D435i RGB frames

Single-slot frame queue

YOLO11n (detection)

Object class + confidence

Two-slot result queue (class, conf.)

non-blocking poll (5 ms)

One-shot strategy scheduler  
First threshold-qualified result (confidence  $\geq 0.25$ )  
object class  $c \rightarrow$  strategy  $s=f(c) \rightarrow$  parameter vector  $\Theta(c)$   
Lock selected strategy for the remainder of the trial

No result or  $\text{con} < 0.25$ :  
keep initial  $\Theta_0$

Unmapped class:  
balanced default

Haptic-interface channel  
 $K_f, d$

Slave-side impedance channel  
 $K_t, K_r, \zeta$

Gripper-execution channel  
 $v_g, F_g$

### *Human-in-the-loop teleoperation supervisor (200 Hz)*

operator command

Operator

Omega.7 master  
(haptic input)

robot command

Panda + Franka Hand  
(impedance / grasp / transport)

Object workspace

In modes C–E, visual inference does not gate operator motion

force & state feedback



asynchronous perception & one-shot updates



continuous teleoperation